

HealthGuard AI - Healthcare Software Requirements Specification

Clinical Data Management & Compliance Automation System

Document Control & Approval Matrix

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Approval Signatures

Role	Name	Signature	Date
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Revision History

Version	Date	Author	Changes
1.0	2025-01-15	Engineering Team	Initial draft
1.5	2025-06-10	Regulatory Team	Added FDA 21 CFR Part 11 compliance
2.0	2025-10-21	Full Team	Comprehensive requirements update

Executive Summary

HealthGuard AI is a comprehensive compliance testing automation platform designed specifically for healthcare software development. This Software Requirements Specification (SRS) document defines the functional, non-functional, data, and interface requirements for HealthGuard AI version 2.0.

The system is designed to meet stringent regulatory requirements including:

- FDA 21 CFR Part 11 (Electronic Records; Electronic Signatures)
- HIPAA Privacy and Security Rules (45 CFR Parts 160, 162, and 164)
- IEC 62304:2006/AMD1:2015 (Medical Device Software Life Cycle Processes)
- ISO 13485:2016 (Medical Devices Quality Management Systems)
- ISO 14971:2019 (Application of Risk Management to Medical Devices)
- GDPR (General Data Protection Regulation) where applicable

This document serves as the primary requirements baseline for system design, development, testing, validation, and regulatory submission.

Target Users:

- Healthcare Software Developers and Quality Assurance Engineers
- Regulatory Affairs Professionals
- Clinical System Administrators
- Healthcare IT Compliance Officers
- FDA and Regulatory Auditors

System Scope: HealthGuard AI provides automated compliance testing, requirements traceability, audit management, and regulatory reporting capabilities for healthcare software systems. The platform integrates with Electronic Health Record (EHR) systems, Laboratory Information Systems (LIS), and Picture Archiving and Communication Systems (PACS).

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Scope & Purpose

3.1 Document Purpose

This Software Requirements Specification (SRS) establishes the complete requirements baseline for HealthGuard AI. It serves multiple critical purposes:

1. **Regulatory Compliance:** Provides evidence of systematic requirements management for FDA 510(k) submission
2. **Development Guidance:** Defines clear, testable requirements for software development teams
3. **Validation Basis:** Establishes acceptance criteria for system validation and verification
4. **Risk Management:** Links requirements to hazard analysis and risk mitigation strategies
5. **Change Control:** Serves as baseline for managing requirements changes throughout the lifecycle

3.2 Product Scope

HealthGuard AI operates within the healthcare software ecosystem as a compliance automation platform. The system boundary includes:

In Scope:

- Automated compliance testing for healthcare applications
- Requirements traceability management
- Audit trail generation and management
- Integration with major EHR systems (Epic, Cerner, MEDITECH)
- HL7 FHIR R4 API compliance testing
- DICOM image handling validation
- Role-based access control with multi-factor authentication
- Secure cloud-based deployment (AWS, Azure, GCP)
- Mobile applications (iOS 15+, Android 12+)

Out of Scope:

- Direct patient care decision support
- Medical device hardware integration (limited to software interfaces)
- Billing and reimbursement processing
- Clinical trial management
- Telemedicine video consultation (video infrastructure only)

3.3 Intended Use Statement

HealthGuard AI is intended for use by healthcare software development teams, quality assurance professionals, and regulatory compliance officers to automate compliance testing, manage requirements traceability, and generate audit documentation for healthcare software systems subject to FDA, HIPAA, and international medical device regulations.

The system is NOT intended for:

- Direct diagnosis or treatment of patients
- Replacement of human clinical judgment
- Use in life-critical scenarios without appropriate human oversight

3.4 User Characteristics

Primary Users:

1. **Healthcare Software Developers** (Experience: 3+ years, Technical expertise: High)
2. **QA/Test Engineers** (Experience: 2+ years, Domain knowledge: Healthcare regulations)
3. **Regulatory Affairs Specialists** (Experience: 5+ years, FDA submission experience)
4. **Clinical System Administrators** (Experience: 2+ years, Healthcare IT background)
5. **Compliance Officers** (Experience: 5+ years, Legal and regulatory expertise)

Training Requirements:

- All users: 8-hour initial training + annual 4-hour refresher

- Administrators: Additional 16-hour advanced administration course
 - Regulatory specialists: 8-hour HealthGuard AI regulatory reporting module
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Regulatory Framework Overview

4.1 FDA 21 CFR Part 11 Compliance

HealthGuard AI complies with FDA 21 CFR Part 11 requirements for electronic records and electronic signatures:

Key Requirements:

- §11.10 - Controls for closed systems (validation, audit trails, authority checks, device checks)
- §11.30 - Controls for open systems (additional security measures)
- §11.50 - Signature manifestations (unique identification, date/time stamping)
- §11.70 - Signature/record linking (cannot be removed, copied, or transferred)
- §11.100 - General requirements for electronic signatures
- §11.200 - Electronic signature components (unique ID, biometrics)
- §11.300 - Controls for identification codes/passwords

4.2 HIPAA Compliance

The system implements comprehensive HIPAA Security and Privacy Rule requirements:

Administrative Safeguards (§164.308):

- Security Management Process (§164.308(a)(1))
- Assigned Security Responsibility (§164.308(a)(2))
- Workforce Security (§164.308(a)(3))
- Information Access Management (§164.308(a)(4))
- Security Awareness and Training (§164.308(a)(5))
- Security Incident Procedures (§164.308(a)(6))
- Contingency Plan (§164.308(a)(7))
- Business Associate Contracts (§164.308(b))

Physical Safeguards (§164.310):

- Facility Access Controls (§164.310(a))
- Workstation Use and Security (§164.310(b), (c))
- Device and Media Controls (§164.310(d))

Technical Safeguards (§164.312):

- Access Control (§164.312(a))
- Audit Controls (§164.312(b))
- Integrity (§164.312(c))
- Person or Entity Authentication (§164.312(d))
- Transmission Security (§164.312(e))

4.3 IEC 62304 Medical Device Software

HealthGuard AI is classified as **Class B software** under IEC 62304:2006/AMD1:2015:

Classification Rationale:

- No injury or damage to health possible → Class A
- **Non-serious injury possible → Class B** ← Applicable

- Death or serious injury possible → Class C

Applicable Requirements:

- Software development planning (§5.1)
- Software requirements analysis (§5.2)
- Software architectural design (§5.3)
- Software detailed design (§5.4)
- Software unit implementation and verification (§5.5)
- Software integration and integration testing (§5.6)
- Software system testing (§5.7)
- Software release (§5.8)

4.4 ISO 13485 Quality Management

The development follows ISO 13485:2016 requirements:

- Design and development planning (§7.3.2)
- Design and development inputs (§7.3.3)
- Design and development outputs (§7.3.4)
- Design and development review (§7.3.5)
- Design and development verification (§7.3.6)
- Design and development validation (§7.3.7)
- Design and development transfer (§7.3.8)
- Control of design and development changes (§7.3.9)

4.5 ISO 14971 Risk Management

Risk management activities per ISO 14971:2019:

- Risk management planning (§4.1)
- Hazard identification (§5.3)
- Risk estimation (§5.4)
- Risk evaluation (§5.5)
- Risk control (§6)
- Residual risk evaluation (§7.1)
- Risk/benefit analysis (§7.2)
- Production and post-production monitoring (§9)

Stakeholder Requirements

5.1 User Stories - Clinical Users

US-001: As a clinician, I need to access patient records securely using multi-factor authentication so that I can ensure HIPAA compliance and protect patient privacy.

US-002: As a physician, I need to view real-time lab results integrated from the LIS so that I can make timely clinical decisions.

US-003: As a nurse, I need to document patient vitals and observations that automatically sync with the EHR so that the medical record is always current.

US-004: As a radiologist, I need to access DICOM images through a PACS-integrated viewer so that I can perform diagnostic interpretations.

US-005: As an emergency physician, I need an emergency access override mechanism so that I can access critical patient data in life-threatening situations.

5.2 User Stories - Administrative Users

US-006: As a system administrator, I need to configure role-based access controls for different user types so that users only have appropriate permissions.

US-007: As a compliance officer, I need to generate comprehensive audit reports filtered by date, user, or action so that I can demonstrate regulatory compliance.

US-008: As a security officer, I need to review failed authentication attempts and potential security incidents so that I can prevent unauthorized access.

US-009: As a quality manager, I need to track requirements traceability from requirements through testing so that I can ensure complete validation coverage.

5.3 User Stories - Regulatory Users

US-010: As a regulatory affairs specialist, I need to generate FDA-ready audit trails with electronic signatures so that I can support 510(k) submissions.

US-011: As a validation engineer, I need to execute automated test scripts against requirements so that I can efficiently validate system functionality.

US-012: As a risk manager, I need to link hazards to mitigating requirements so that I can demonstrate risk control effectiveness.

Functional Requirements

6.1 User Management & Authentication

REQ-AUTH-001: Multi-Factor Authentication (MFA)

Title: Multi-Factor Authentication for All Users

Description: The system shall require multi-factor authentication for all user accounts accessing patient health information (PHI) or system administration functions. MFA must combine at least two of the following factors: knowledge factor (password), possession factor (mobile device, security token), or inherence factor (biometric). The implementation must support multiple MFA methods including SMS OTP, authenticator apps (TOTP), hardware tokens (FIDO2), and biometric authentication.

Priority: Critical

Category: Functional - Security

Compliance Standards:

- HIPAA §164.312(a)(2)(i) - Access Control - Authentication
- HIPAA §164.308(a)(5)(ii)(D) - Password Management
- FDA 21 CFR Part 11.300(b) - Password Requirements
- NIST SP 800-63B - Digital Identity Guidelines
- ISO/IEC 27001:2013 - A.9.4.2 Secure Authentication

Risk Level: High (Mitigates: Unauthorized access to PHI)

Rationale: MFA significantly reduces the risk of unauthorized access due to compromised credentials. Healthcare data breaches often result from weak or stolen passwords. This requirement ensures strong authentication aligned with current cybersecurity best practices and regulatory expectations.

Acceptance Criteria:

1. The system shall prompt users to configure MFA during initial account setup within 24 hours of account creation.
2. Users shall not be able to access PHI without completing MFA setup after the grace period expires.
3. The system shall support minimum three MFA methods: SMS OTP, authenticator app (TOTP), and biometric (fingerprint/face).
4. MFA challenges shall occur at initial login and after session timeout (60 minutes of inactivity).
5. The system shall allow administrators to enforce MFA policies including requiring re-authentication for sensitive operations.
6. Failed MFA attempts shall be logged with user ID, timestamp, IP address, and attempt type.
7. After 5 consecutive failed MFA attempts, the account shall be temporarily locked for 30 minutes.
8. The system shall provide backup recovery codes (minimum 10 single-use codes) during MFA setup.
9. MFA tokens shall expire after 30 seconds for TOTP and 5 minutes for SMS OTP.
10. The system shall support emergency access bypass for designated clinical roles with mandatory post-access audit.

Dependencies:

- REQ-AUTH-002 (Password Policy)
- REQ-AUTH-003 (Session Management)
- REQ-AUDIT-001 (Authentication Event Logging)
- REQ-DATA-005 (Encryption in Transit)

Test Cases:

1. **TC-AUTH-001-01:** Verify new user receives MFA setup prompt on first login
2. **TC-AUTH-001-02:** Verify access denied without MFA completion after grace period
3. **TC-AUTH-001-03:** Verify successful login with valid TOTP code
4. **TC-AUTH-001-04:** Verify failed login with invalid TOTP code and proper error message
5. **TC-AUTH-001-05:** Verify account lockout after 5 failed MFA attempts
6. **TC-AUTH-001-06:** Verify emergency access override works with proper audit logging
7. **TC-AUTH-001-07:** Verify backup recovery codes work and are single-use only
8. **TC-AUTH-001-08:** Verify expired TOTP codes are rejected
9. **TC-AUTH-001-09:** Verify biometric authentication on supported devices
10. **TC-AUTH-001-10:** Negative test - Verify replay attack protection

Validation Method: Testing (Automated + Manual), Inspection (Code Review)

Traceability:

- Design Spec: DS-AUTH-001
- Test Plan: TP-SEC-001
- Risk Analysis: HA-001 (Unauthorized Access)
- User Story: US-001

REQ-AUTH-002: Password Policy Enforcement

Title: Comprehensive Password Security Policy

Description: The system shall enforce a comprehensive password policy compliant with NIST SP 800-63B guidelines and HIPAA security requirements. Passwords must meet complexity requirements, length minimums, and rotation policies. The system must detect and prevent use of compromised passwords using breach databases and implement secure password storage using industry-standard hashing algorithms.

Priority: Critical

Category: Functional - Security

Compliance Standards:

- HIPAA §164.308(a)(5)(ii)(D) - Password Management
- FDA 21 CFR Part 11.300(a) - Password Controls
- NIST SP 800-63B §5.1.1 - Memorized Secret Verifiers
- ISO/IEC 27001 A.9.4.3 - Password Management System
- PCI DSS 8.2 - Password Complexity

Risk Level: High

Rationale: Weak passwords remain a primary attack vector for healthcare data breaches. Strong password policies reduce brute force attack success and credential compromise risks.

Acceptance Criteria:

1. Passwords shall be minimum 12 characters in length for standard users and 16 characters for administrators.
2. Passwords shall require at least three of four character types: uppercase, lowercase, numbers, special characters (!@#\$%^&*).
3. The system shall reject passwords matching the user's username, email, or display name (case-insensitive).
4. The system shall check passwords against Have I Been Pwned API (or equivalent) and reject compromised passwords.
5. Password history shall prevent reuse of the last 24 passwords.
6. Password expiration shall be enforced every 90 days for standard users and 60 days for administrators.
7. The system shall provide password strength indicator in real-time during password creation.
8. Passwords shall be stored using bcrypt with minimum work factor of 12 (or Argon2id with equivalent security).
9. The system shall not transmit passwords in clear text or store them reversibly encrypted.
10. Users shall receive email notification 14 days and 7 days before password expiration.
11. Temporary passwords issued by administrators shall expire after 24 hours and force immediate change.
12. The system shall lock accounts after 5 consecutive failed password attempts for 30 minutes.

Dependencies:

- REQ-AUTH-001 (Multi-Factor Authentication)
- REQ-AUTH-005 (Account Lockout)
- REQ-AUDIT-002 (Failed Login Logging)
- REQ-DATA-001 (Data Encryption at Rest)

Test Cases:

1. **TC-AUTH-002-01:** Verify password length validation (accept ≥ 12 chars, reject < 12 chars)
2. **TC-AUTH-002-02:** Verify complexity requirements (test all combinations)
3. **TC-AUTH-002-03:** Verify rejection of compromised passwords from breach database
4. **TC-AUTH-002-04:** Verify password history prevents reuse of last 24 passwords
5. **TC-AUTH-002-05:** Verify password expiration after 90 days
6. **TC-AUTH-002-06:** Verify bcrypt hashing with work factor ≥ 12

- 7. **TC-AUTH-002-07:** Verify temporary password expiration after 24 hours
- 8. **TC-AUTH-002-08:** Verify account lockout after 5 failed attempts
- 9. **TC-AUTH-002-09:** Negative test - Verify plain text passwords are never stored or logged
- 10. **TC-AUTH-002-10:** Performance test - Verify password hash computation time is 100-500ms

Validation Method: Testing (Automated), Analysis (Static Code Analysis), Inspection (Security Review)

Traceability:

- Design Spec: DS-AUTH-002
- Test Plan: TP-SEC-002
- Risk Analysis: HA-001, HA-003
- User Story: US-001

REQ-AUTH-003: Role-Based Access Control (RBAC)

Title: Comprehensive Role-Based Access Control System

Description: The system shall implement a comprehensive role-based access control (RBAC) framework supporting minimum five predefined roles with granular permissions. Access rights shall be assigned based on job function and clinical responsibilities, implementing least privilege principle. The system must support role hierarchies, role delegation, and dynamic permission management with full audit trail.

Priority: Critical

Category: Functional - Security & Access Control

Compliance Standards:

- HIPAA §164.308(a)(4) - Information Access Management
- HIPAA §164.308(a)(3) - Workforce Security
- HIPAA §164.312(a)(1) - Access Control
- ISO/IEC 27001 A.9.1.2 - Access to Networks and Network Services
- NIST SP 800-53 AC-2 - Account Management

Risk Level: High

Rationale: Healthcare organizations require complex access controls to ensure users only access PHI necessary for their job functions. RBAC reduces administrative overhead while maintaining strict access governance required by HIPAA minimum necessary standard.

Predefined Roles:

1. **System Administrator:** Full system configuration, user management, security settings
2. **Physician:** Full patient data access, order entry, clinical documentation
3. **Nurse:** Patient data view, vital signs entry, medication administration
4. **Lab Technician:** Lab orders view, result entry, specimen tracking
5. **Billing Specialist:** Patient demographics, insurance info, billing codes (no clinical data)
6. **Regulatory Auditor:** Read-only access to audit logs and compliance reports
7. **Research Coordinator:** De-identified data access only

Acceptance Criteria:

1. The system shall support minimum 7 predefined roles with distinct permission sets as specified above.
2. Each user account shall be assigned to one or more roles, with permissions being union of all assigned roles.
3. Role permissions shall include granular controls: Create, Read, Update, Delete, Print, Export for each data category.

4. The system shall enforce role-based data filtering ensuring users only see data within their scope.
5. Administrators shall be able to create custom roles by combining existing permission sets.
6. Role assignments and changes shall require approval from two authorized administrators.
7. All role assignments, modifications, and deletions shall be logged in tamper-proof audit trail.
8. The system shall support temporary role elevation for maximum 24 hours with automatic expiration.
9. Role permissions shall be validated on every data access request (not just at login).
10. The system shall display current user role and permissions on user profile page.
11. Physicians shall have "break glass" emergency access capability with mandatory post-access justification.
12. Users shall receive email notification when their role permissions are modified.
13. The system shall enforce separation of duties: no single user can create and approve critical operations.

Dependencies:

- REQ-AUTH-001 (Multi-Factor Authentication)
- REQ-AUDIT-001 (Authentication Event Logging)
- REQ-AUDIT-003 (Data Access Logging)
- REQ-DATA-003 (Data Segmentation)

Test Cases:

1. **TC-AUTH-003-01:** Verify physician can access full patient chart
2. **TC-AUTH-003-02:** Verify nurse cannot access billing information
3. **TC-AUTH-003-03:** Verify billing specialist cannot access clinical notes
4. **TC-AUTH-003-04:** Verify lab technician can only access lab-related data
5. **TC-AUTH-003-05:** Verify regulatory auditor has read-only access to audit logs
6. **TC-AUTH-003-06:** Verify role changes require two-person approval
7. **TC-AUTH-003-07:** Verify temporary role elevation expires after 24 hours
8. **TC-AUTH-003-08:** Verify "break glass" emergency access logs properly
9. **TC-AUTH-003-09:** Verify permissions validated on each API request, not cached
10. **TC-AUTH-003-10:** Negative test - Verify privilege escalation attacks are blocked
11. **TC-AUTH-003-11:** Performance test - Verify permission check latency <50ms

Validation Method: Testing (Automated + Penetration), Demonstration (Security Review)

Traceability:

- Design Spec: DS-AUTH-003
- Test Plan: TP-SEC-003, TP-RBAC-001
- Risk Analysis: HA-002 (Unauthorized Data Access)
- User Story: US-006

REQ-AUTH-004: Session Management & Timeout

Title: Secure Session Management with Automatic Timeout

Description: The system shall implement secure session management including automatic timeout, session invalidation, and protection against session hijacking attacks. Sessions must time out after periods of inactivity to prevent unauthorized access from unattended workstations, with configurable timeout periods based on data sensitivity and user role.

Priority: High

Category: Functional - Security

Compliance Standards:

- HIPAA §164.312(a)(2)(iii) - Automatic Logoff
- NIST SP 800-53 AC-12 - Session Termination
- OWASP Session Management Cheat Sheet
- PCI DSS 8.1.8 - Session Timeout

Risk Level: Medium

Rationale: Healthcare workstations are frequently left unattended in clinical environments. Automatic session timeout prevents unauthorized individuals from accessing active sessions containing PHI.

Acceptance Criteria:

1. Sessions shall automatically timeout after 15 minutes of user inactivity for workstations accessing PHI.
2. Administrative sessions shall timeout after 10 minutes of inactivity.
3. The system shall display a countdown warning 2 minutes before session expiration.
4. Users shall be able to extend their session by clicking "Stay Logged In" during warning period.
5. Session tokens shall be cryptographically random, minimum 128 bits entropy.
6. Session tokens shall be stored using HTTP-only, Secure, and SameSite=Strict cookie flags.
7. The system shall invalidate session tokens immediately upon user logout.
8. A user's prior sessions shall be automatically terminated when they authenticate from a new device.
9. The system shall limit concurrent sessions to 3 per user account.
10. Session tokens shall rotate after successful authentication and privilege elevation.
11. The system shall bind sessions to client IP address and User-Agent to prevent session hijacking.
12. All session events (creation, timeout, termination) shall be logged with timestamp and client information.

Dependencies:

- REQ-AUTH-001 (Multi-Factor Authentication)
- REQ-AUDIT-001 (Authentication Event Logging)
- REQ-DATA-005 (Encryption in Transit)

Test Cases:

1. **TC-AUTH-004-01:** Verify session times out after exactly 15 minutes of inactivity
2. **TC-AUTH-004-02:** Verify 2-minute warning appears before timeout
3. **TC-AUTH-004-03:** Verify "Stay Logged In" extends session properly
4. **TC-AUTH-004-04:** Verify session invalidated on user logout
5. **TC-AUTH-004-05:** Verify session token randomness and entropy
6. **TC-AUTH-004-06:** Verify HTTP-only and Secure flags on session cookies
7. **TC-AUTH-004-07:** Verify concurrent session limit enforced (max 3 sessions)
8. **TC-AUTH-004-08:** Verify session bound to IP address and User-Agent
9. **TC-AUTH-004-09:** Negative test - Verify stolen session token cannot be reused from different IP
10. **TC-AUTH-004-10:** Negative test - Verify session fixation attack prevention

Validation Method: Testing (Automated + Penetration Testing)

Traceability:

- Design Spec: DS-AUTH-004
- Test Plan: TP-SEC-004
- Risk Analysis: HA-004 (Session Hijacking)
- User Story: US-001

REQ-AUTH-005: Single Sign-On (SSO) Integration

Title: Enterprise SSO Integration via SAML 2.0 and OAuth 2.0

Description: The system shall support enterprise Single Sign-On (SSO) integration using industry-standard protocols SAML 2.0 and OAuth 2.0/OpenID Connect. SSO integration enables healthcare organizations to centralize authentication management, reduce password fatigue, and improve security through integration with existing identity providers such as Active Directory, Okta, Azure AD, and Auth0.

Priority: High

Category: Functional - Integration & Security

Compliance Standards:

- NIST SP 800-63C - Federation and Assertions
- HIPAA §164.312(d) - Person or Entity Authentication
- SAML 2.0 Core Specification (OASIS Standard)
- OAuth 2.0 Authorization Framework (RFC 6749)
- OpenID Connect Core 1.0

Risk Level: Medium

Rationale: Healthcare enterprises typically have existing identity management infrastructure. SSO integration reduces authentication friction, improves user experience, and centralizes security policy enforcement across enterprise applications.

Acceptance Criteria:

1. The system shall support SAML 2.0 Service Provider (SP) initiated and Identity Provider (IdP) initiated flows.
2. The system shall validate SAML assertions including signature verification, timestamp validation, and audience restriction.
3. The system shall support OAuth 2.0 Authorization Code Grant with PKCE (Proof Key for Code Exchange).
4. The system shall implement OpenID Connect for user profile information retrieval.
5. SSO configuration shall support multiple identity providers simultaneously (multi-tenancy).
6. The system shall extract user roles and permissions from SAML attributes or OAuth claims.
7. Just-In-Time (JIT) user provisioning shall create user accounts automatically on first SSO login.
8. The system shall provide SSO metadata endpoint for IdP configuration.
9. SSO sessions shall respect MFA requirements enforced by the IdP.
10. The system shall log all SSO authentication events including IdP name, assertion ID, and user claims.
11. SSO logout shall trigger both local logout and IdP logout (Single Logout - SLO).
12. The system shall fall back to local authentication if SSO provider is unavailable.

Dependencies:

- REQ-AUTH-001 (Multi-Factor Authentication)
- REQ-AUTH-003 (Role-Based Access Control)
- REQ-AUDIT-001 (Authentication Event Logging)

Test Cases:

1. **TC-AUTH-005-01:** Verify successful SAML SP-initiated SSO flow
2. **TC-AUTH-005-02:** Verify successful SAML IdP-initiated SSO flow
3. **TC-AUTH-005-03:** Verify OAuth 2.0 authorization code flow with PKCE
4. **TC-AUTH-005-04:** Verify OpenID Connect user info retrieval
5. **TC-AUTH-005-05:** Verify SAML assertion signature validation
6. **TC-AUTH-005-06:** Verify role mapping from SAML attributes to internal roles
7. **TC-AUTH-005-07:** Verify JIT user provisioning creates account on first login
8. **TC-AUTH-005-08:** Verify Single Logout (SLO) terminates both local and IdP sessions

- 9. **TC-AUTH-005-09:** Verify fallback to local auth when IdP unavailable
- 10. **TC-AUTH-005-10:** Negative test - Verify tampered SAML assertion is rejected
- 11. **TC-AUTH-005-11:** Security test - Verify protection against XML signature wrapping attacks

Validation Method: Testing (Automated + Integration Testing), Demonstration

Traceability:

- Design Spec: DS-AUTH-005, DS-INT-003
 - Test Plan: TP-SEC-005, TP-INT-002
 - Risk Analysis: HA-005 (SSO Configuration Errors)
 - User Story: US-001, US-006
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6.2 Patient Data Management

REQ-DATA-001: Patient Data Encryption at Rest

Title: AES-256 Encryption for All PHI at Rest

Description: The system shall encrypt all Protected Health Information (PHI) and Personally Identifiable Information (PII) at rest using AES-256-GCM encryption algorithm. Encryption shall be applied at the database level, file system level, and backup storage. Encryption keys must be managed using industry-standard key management services (KMS) with key rotation, access controls, and audit logging.

Priority: Critical

Category: Data Security

Compliance Standards:

- HIPAA §164.312(a)(2)(iv) - Encryption and Decryption
- HIPAA §164.312(e)(2)(ii) - Encryption
- FDA 21 CFR Part 11.10(a) - Validation of Systems
- NIST SP 800-111 - Storage Encryption Technologies
- NIST FIPS 140-2 - Cryptographic Module Validation
- GDPR Article 32 - Security of Processing

Risk Level: Critical

Rationale: Healthcare data breaches frequently occur due to stolen or improperly disposed storage media. Encryption at rest ensures that data remains protected even if physical security is compromised. This is a foundational control for HIPAA Security Rule compliance.

Acceptance Criteria:

1. All database columns containing PHI/PII shall be encrypted using AES-256-GCM with authenticated encryption.
2. Database encryption keys shall be stored in AWS KMS, Azure Key Vault, or GCP Cloud KMS (not in application code or config files).
3. Encryption keys shall be unique per customer/tenant in multi-tenant deployments.
4. Automatic key rotation shall occur every 365 days with re-encryption of data using new keys.
5. File uploads containing PHI (documents, images, PDFs) shall be encrypted immediately upon storage.
6. Database backups shall be encrypted using separate backup encryption keys.
7. Temporary files and cache containing PHI shall be encrypted or securely wiped after use.
8. Encryption key access shall be restricted to authorized service accounts only, with all access logged.
9. The system shall use hardware security modules (HSM) for key storage when available (FIPS 140-2 Level 2+).

10. Encryption shall have negligible impact on system performance (<5% overhead).
11. Disaster recovery procedures shall include secure key backup and recovery mechanisms.
12. The system shall provide encryption status dashboard showing encryption coverage percentage.

Dependencies:

- REQ-DATA-005 (Encryption in Transit)
- REQ-AUDIT-006 (Encryption Key Access Logging)
- REQ-SYS-008 (Backup and Recovery)

Test Cases:

1. **TC-DATA-001-01:** Verify all PHI database columns are encrypted at rest
2. **TC-DATA-001-02:** Verify AES-256-GCM algorithm is used for encryption
3. **TC-DATA-001-03:** Verify encryption keys stored in KMS, not in code/config
4. **TC-DATA-001-04:** Verify key rotation occurs every 365 days
5. **TC-DATA-001-05:** Verify uploaded files are encrypted upon storage
6. **TC-DATA-001-06:** Verify database backups are encrypted
7. **TC-DATA-001-07:** Verify encryption key access is restricted and logged
8. **TC-DATA-001-08:** Performance test - Verify encryption overhead <5%
9. **TC-DATA-001-09:** Verify disaster recovery key restoration process
10. **TC-DATA-001-10:** Negative test - Verify data unreadable without encryption keys
11. **TC-DATA-001-11:** Security test - Verify keys cannot be extracted from system

Validation Method: Testing (Automated), Analysis (Cryptographic Review), Inspection (Security Audit)

Traceability:

- Design Spec: DS-DATA-001, DS-SEC-002
- Test Plan: TP-SEC-006, TP-DATA-001
- Risk Analysis: HA-006 (Data Breach), HA-007 (Media Theft)
- User Story: US-001

REQ-DATA-002: Patient Health Record CRUD Operations

Title: Comprehensive Patient Health Record Management

Description: The system shall provide complete Create, Read, Update, and Delete (CRUD) operations for patient health records with workflow validation, data integrity checks, version control, and comprehensive audit trails. All operations must comply with HL7 FHIR R4 data models and support concurrent access with optimistic locking to prevent data conflicts.

Priority: Critical

Category: Functional - Data Management

Compliance Standards:

- HIPAA §164.308(a)(4)(ii)(C) - Access Establishment and Modification
- HL7 FHIR R4 Specification
- ICD-10-CM Diagnosis Codes
- CPT/HCPCS Procedure Codes
- SNOMED CT Clinical Terminology
- LOINC Lab Observation Codes

Risk Level: High

Rationale: Patient health records are the core data asset in healthcare systems. Proper CRUD operations ensure data integrity, support clinical workflows, and maintain comprehensive audit trails required for regulatory compliance and medical-legal documentation.

Acceptance Criteria:

Create Operations:

1. New patient records shall validate required fields: MRN, Name, DOB, Gender, Contact Info.
2. Medical Record Number (MRN) shall be system-generated using configurable format (e.g., HG-YYYYMMDD-NNNN).
3. Duplicate patient detection shall check for existing records matching Name+DOB+SSN with 95% confidence score.
4. Patient demographics shall support HL7 FHIR Patient resource structure.
5. Initial record creation shall capture creating user, timestamp, and facility/location.

Read Operations: 6. Patient search shall support multiple criteria: MRN, Name, DOB, SSN, Phone, Email. 7. Search results shall be paginated (50 records per page) with total count display. 8. Patient chart view shall aggregate data from multiple sources: demographics, encounters, diagnoses, medications, labs, imaging. 9. Data retrieval shall enforce role-based filtering (nurses see nursing notes, physicians see full chart). 10. The system shall log every PHI access with purpose of use (Treatment, Payment, Operations).

Update Operations: 11. Patient demographic updates shall require reason for change documentation. 12. Clinical data updates shall create new versions while preserving prior versions (append-only audit log). 13. Concurrent updates shall use optimistic locking with version numbers to prevent conflicts. 14. If concurrent update conflict detected, system shall alert user and require manual conflict resolution. 15. Medication orders shall require approval workflow: Ordered → Verified → Administered → Documented. 16. Critical field updates (Name, DOB, MRN) shall require supervisor approval.

Delete Operations: 17. Patient records shall NEVER be physically deleted (hard delete) from the database. 18. Logical deletion (soft delete) shall mark records as inactive/archived with deletion reason. 19. Deleted records shall remain in database for retention period (7 years minimum per HIPAA). 20. Only System Administrators can mark records as deleted, with mandatory justification.

Data Integrity: 21. All diagnosis codes shall validate against ICD-10-CM code set with version date. 22. Procedure codes shall validate against CPT/HCPCS code sets. 23. Lab results shall use LOINC codes for test identification. 24. Medication names shall validate against RxNorm database. 25. All date fields shall validate for reasonable ranges (DOB not in future, etc.).

Dependencies:

- REQ-AUTH-003 (Role-Based Access Control)
- REQ-DATA-001 (Encryption at Rest)
- REQ-AUDIT-003 (Data Access Logging)
- REQ-INT-001 (HL7 FHIR API)

Test Cases:

1. **TC-DATA-002-01:** Verify patient creation with all required fields
2. **TC-DATA-002-02:** Verify MRN auto-generation with correct format
3. **TC-DATA-002-03:** Verify duplicate patient detection
4. **TC-DATA-002-04:** Verify patient search by multiple criteria
5. **TC-DATA-002-05:** Verify role-based data filtering in patient chart
6. **TC-DATA-002-06:** Verify demographic update with reason documentation
7. **TC-DATA-002-07:** Verify version control creates new versions

8. **TC-DATA-002-08:** Verify optimistic locking prevents concurrent update conflicts
9. **TC-DATA-002-09:** Verify medication order approval workflow
10. **TC-DATA-002-10:** Verify soft delete instead of hard delete
11. **TC-DATA-002-11:** Verify ICD-10-CM code validation
12. **TC-DATA-002-12:** Verify LOINC code validation for lab results
13. **TC-DATA-002-13:** Performance test - Verify patient search response time <2 seconds
14. **TC-DATA-002-14:** Negative test - Verify invalid ICD-10 codes are rejected
15. **TC-DATA-002-15:** Stress test - Verify system handles 1000 concurrent patient chart accesses

Validation Method: Testing (Automated + Manual), Demonstration (Workflow Review)

Traceability:

- Design Spec: DS-DATA-002, DS-WORKFLOW-001
 - Test Plan: TP-FUNC-001, TP-DATA-002
 - Risk Analysis: HA-008 (Data Integrity Errors), HA-009 (Concurrent Access Conflicts)
 - User Story: US-002, US-003
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REQ-DATA-003: Data De-identification and Anonymization

Title: HIPAA-Compliant Data De-identification for Research and Analytics

Description: The system shall provide automated de-identification and anonymization capabilities compliant with HIPAA Safe Harbor and Expert Determination methods. This enables use of healthcare data for research, quality improvement, and analytics while protecting patient privacy. The system must remove or transform all 18 HIPAA identifiers and provide cryptographic hashing for re-identification when authorized.

Priority: High

Category: Functional - Data Privacy

Compliance Standards:

- HIPAA §164.514(a) - De-identification Standard
- HIPAA §164.514(b) - Safe Harbor Method
- HIPAA §164.514(c) - Expert Determination Method
- GDPR Article 89 - Safeguards for Processing for Research Purposes
- FDA Guidance on De-identification of Medical Device Data

Risk Level: High

Rationale: Healthcare organizations need to use clinical data for research, quality improvement, and machine learning while protecting patient privacy. De-identification enables secondary use of data in compliance with HIPAA Privacy Rule.

18 HIPAA Identifiers to Remove:

1. Names
2. Geographic subdivisions smaller than state
3. Dates (except year) related to individual
4. Telephone numbers
5. Fax numbers
6. Email addresses
7. Social Security numbers
8. Medical record numbers
9. Health plan beneficiary numbers

10. Account numbers
11. Certificate/license numbers
12. Vehicle identifiers and serial numbers
13. Device identifiers and serial numbers
14. Web URLs
15. IP addresses
16. Biometric identifiers
17. Full-face photos
18. Any other unique identifying numbers/characteristics

Acceptance Criteria:

1. The system shall provide "De-identify Dataset" function accessible to authorized researchers and analysts.
2. De-identification shall remove all 18 HIPAA identifiers using Safe Harbor method by default.
3. Names shall be replaced with pseudonyms (Patient-001, Patient-002) maintaining consistency within dataset.
4. Dates shall be shifted by random offset (± 30 days) while preserving time intervals.
5. Geographic locations shall be generalized to state level or three-digit ZIP codes.
6. Medical record numbers shall be replaced with cryptographic hashes (SHA-256) for linkage.
7. Ages ≥ 90 shall be aggregated to "90+" category.
8. Rare diagnoses/procedures appearing in < 5 records shall be generalized to higher-level codes.
9. The system shall provide "Expert Determination" mode allowing custom de-identification rules.
10. De-identification process shall be logged with dataset ID, method used, and performing user.
11. Re-identification keys shall be stored separately with restricted access requiring dual authorization.
12. The system shall perform statistical disclosure risk analysis calculating k-anonymity and l-diversity metrics.
13. De-identified datasets shall include metadata documenting de-identification method and date.
14. The system shall support differential privacy techniques for aggregate statistics with configurable epsilon values.

Dependencies:

- REQ-AUTH-003 (Role-Based Access Control)
- REQ-DATA-001 (Encryption at Rest)
- REQ-AUDIT-004 (De-identification Event Logging)
- REQ-DATA-008 (Data Retention)

Test Cases:

1. **TC-DATA-003-01:** Verify all 18 HIPAA identifiers removed in Safe Harbor mode
2. **TC-DATA-003-02:** Verify names replaced with consistent pseudonyms
3. **TC-DATA-003-03:** Verify date shifting maintains time intervals correctly
4. **TC-DATA-003-04:** Verify ZIP codes truncated to 3 digits
5. **TC-DATA-003-05:** Verify ages ≥ 90 aggregated to "90+"
6. **TC-DATA-003-06:** Verify rare diagnoses generalized (occurring < 5 times)
7. **TC-DATA-003-07:** Verify k-anonymity calculation with $k \geq 5$
8. **TC-DATA-003-08:** Verify re-identification requires dual authorization
9. **TC-DATA-003-09:** Verify de-identification metadata included in export
10. **TC-DATA-003-10:** Negative test - Verify re-identification keys cannot be accessed by researchers
11. **TC-DATA-003-11:** Validation test - Manual review of 100 de-identified records for residual identifiers

Validation Method: Testing (Automated), Analysis (Statistical Disclosure Risk), Inspection (Privacy Expert Review)

Traceability:

- Design Spec: DS-DATA-003, DS-PRIVACY-001

- Test Plan: TP-PRIV-001, TP-DATA-003
 - Risk Analysis: HA-010 (Re-identification Risk)
 - User Story: US-009
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6.3 Clinical Workflows

REQ-CLIN-001: Electronic Health Record (EHR) Integration

Title: HL7 FHIR R4 Integration with Major EHR Systems

Description: The system shall integrate with major Electronic Health Record (EHR) systems including Epic, Cerner Oracle Health, MEDITECH, and Allscripts using HL7 FHIR R4 API standards. Integration shall enable bidirectional data exchange including patient demographics, encounters, diagnoses, medications, lab results, and clinical notes. The system must support real-time data synchronization with conflict resolution and error handling.

Priority: Critical

Category: Interface - Clinical Integration

Compliance Standards:

- HL7 FHIR R4 Specification
- ONC 21st Century Cures Act - Information Blocking
- CMS Interoperability and Patient Access Final Rule
- HL7 Version 2.x (v2.5.1, v2.7)
- HITRUST CSF v10 - Information Exchange

Risk Level: High

Rationale: Healthcare providers use diverse EHR systems. Seamless integration is essential for clinical workflow adoption, data completeness, and eliminating manual data entry errors. FHIR R4 provides standardized interoperability across vendor platforms.

Supported EHR Systems:

- Epic (FHIR R4 API via App Orchard)
- Cerner Oracle Health (FHIR R4 via Code Console)
- MEDITECH (FHIR R4 API)
- Allscripts (FHIR R4)
- Generic SMART-on-FHIR compliant systems

Acceptance Criteria:

Authentication & Authorization:

1. The system shall support SMART-on-FHIR authorization using OAuth 2.0.
2. EHR integration shall use client credentials or authorization code grant flows.
3. Access tokens shall be cached and automatically refreshed before expiration.
4. Token refresh failures shall trigger administrator alerts.

Patient Demographics Synchronization: 5. Patient demographics (FHIR Patient resource) shall sync bidirectionally every 5 minutes. 6. Conflicting updates shall use "latest timestamp wins" with conflict log for manual review. 7. New patients created in EHR shall auto-create corresponding records in HealthGuard AI.

Clinical Data Exchange: 8. The system shall retrieve FHIR resources: Patient, Encounter, Condition, Procedure, Observation, MedicationRequest, DiagnosticReport, DocumentReference. 9. Lab results (FHIR Observation) shall sync every 10 minutes with LOINC code mapping. 10. Medication orders (FHIR MedicationRequest) shall sync in real-time

(<30 seconds). 11. Clinical notes (FHIR DocumentReference) shall be retrieved on-demand with full text content. 12. Diagnostic imaging orders shall sync with links to PACS viewer.

Data Mapping & Validation: 13. ICD-10-CM codes shall map between EHR and internal formats with version validation. 14. SNOMED CT codes shall be preserved during data exchange. 15. LOINC codes for lab tests shall be validated against current LOINC database. 16. RxNorm codes for medications shall be validated and mapped to internal formulary.

Error Handling & Reliability: 17. API failures shall implement exponential backoff retry (max 5 attempts, up to 5 minutes). 18. Failed synchronizations shall queue for manual resolution with admin notification. 19. The system shall maintain integration health dashboard showing success/failure rates per EHR. 20. Integration errors shall be categorized: Authentication, Network, Mapping, Validation.

Performance & Scalability: 21. Patient search via EHR API shall return results within 3 seconds. 22. Bulk data retrieval shall support 1000 patients per batch with progress tracking. 23. The system shall support rate limiting per EHR vendor specifications (Epic: 1000 req/5min).

Dependencies:

- REQ-INT-001 (HL7 FHIR API Implementation)
- REQ-DATA-002 (Patient Data CRUD)
- REQ-AUDIT-005 (EHR Data Exchange Logging)
- REQ-SYS-002 (System Availability)

Test Cases:

1. **TC-CLIN-001-01:** Verify SMART-on-FHIR OAuth authentication with Epic sandbox
2. **TC-CLIN-001-02:** Verify patient demographics sync from Epic to HealthGuard AI
3. **TC-CLIN-001-03:** Verify lab results (FHIR Observation) retrieval with LOINC codes
4. **TC-CLIN-001-04:** Verify medication orders (FHIR MedicationRequest) real-time sync
5. **TC-CLIN-001-05:** Verify conflict resolution using latest timestamp
6. **TC-CLIN-001-06:** Verify ICD-10-CM code mapping accuracy
7. **TC-CLIN-001-07:** Verify exponential backoff on API failure
8. **TC-CLIN-001-08:** Verify integration health dashboard accuracy
9. **TC-CLIN-001-09:** Performance test - Verify patient search response <3 seconds
10. **TC-CLIN-001-10:** Integration test - End-to-end patient flow from EHR to HealthGuard AI
11. **TC-CLIN-001-11:** Negative test - Verify graceful handling of invalid FHIR resources
12. **TC-CLIN-001-12:** Load test - Verify handling of 1000 patient bulk sync

Validation Method: Testing (Integration Testing with EHR Sandboxes), Demonstration

Traceability:

- Design Spec: DS-INT-001, DS-FHIR-001
- Test Plan: TP-INT-001, TP-CLIN-001
- Risk Analysis: HA-011 (Integration Failures), HA-012 (Data Mapping Errors)
- User Story: US-002, US-007

REQ-CLIN-002: Clinical Decision Support Alerts

Title: Rule-Based Clinical Decision Support System

Description: The system shall provide a rule-based Clinical Decision Support (CDS) system that evaluates patient data against clinical rules and triggers alerts for drug interactions, allergy contraindications, duplicate therapy, critical

lab values, and evidence-based care recommendations. CDS alerts must be configurable by clinical administrators and support alert fatigue mitigation through intelligent prioritization.

Priority: High

Category: Functional - Clinical Safety

Compliance Standards:

- FDA Guidance on Clinical Decision Support Software
- IEC 62304 Class B Software Safety Requirements
- ISO 14971 Risk Management (Patient Harm Hazards)
- CMS Merit-based Incentive Payment System (MIPS) CDS Credit
- HITRUST CSF - Clinical Decision Support

Risk Level: Critical (Mitigates patient safety risks)

Rationale: Clinical decision support reduces medical errors, improves care quality, and prevents adverse drug events. Healthcare systems require intelligent alerting that balances safety with usability to avoid alert fatigue and alert override behavior.

CDS Rule Categories:

1. **Drug-Drug Interactions:** Major interactions requiring intervention
2. **Drug-Allergy Contraindications:** Medications contraindicated by documented allergies
3. **Duplicate Therapy:** Multiple medications from same therapeutic class
4. **Dosing Alerts:** Dose exceeds recommended maximum
5. **Lab Value Alerts:** Critical lab results (panic values)
6. **Preventive Care Reminders:** Overdue screenings/vaccinations
7. **Evidence-Based Guidelines:** Protocol deviations

Acceptance Criteria:

Alert Generation:

1. Drug interaction checking shall use First Databank or similar commercial interaction database.
2. Interaction alerts shall classify severity: Critical (stop order), Major (require override), Moderate (informational).
3. Allergy checks shall trigger immediately when medication order entered for patient with documented allergy.
4. Critical lab alerts (panic values) shall notify ordering physician within 1 minute of result entry.
5. Duplicate therapy alerts shall detect same therapeutic class using RxNorm normalized codes.

Alert Display & Workflow: 6. Critical alerts shall be modal dialogs requiring explicit acknowledgment before proceeding. 7. Alert dialogs shall display: Patient name, alert description, clinical evidence, recommended action. 8. Users shall be able to override alerts by selecting reason: Benefit outweighs risk, Patient request, Other (free text required). 9. All alert presentations and user responses shall be logged for analysis. 10. Alert override rates shall be monitored per provider with threshold alerts for >20% override rate.

Alert Fatigue Mitigation: 11. Similar alerts for same patient shall be deduplicated and shown once per session. 12. Informational alerts shall be minimized; only actionable alerts shown by default. 13. Alert priority scoring shall consider patient risk factors (age, renal function, pregnancy). 14. Users shall be able to dismiss low-priority alerts to "Defer to Pharmacist Review." 15. The system shall learn from user behavior and suggest alert threshold adjustments.

Configurable Rules: 16. Clinical administrators shall be able to create custom CDS rules using web-based rule editor. 17. Rule conditions shall support: Age, Gender, Diagnosis codes, Lab values, Medication orders, Vital signs. 18. Rules

shall be versioned with effective dates and change history. 19. New rules shall be tested in "monitor mode" before active enforcement. 20. Rule effectiveness metrics (alert acceptance rate, outcome impact) shall be tracked.

Performance: 21. CDS rule evaluation shall occur within 500ms of order entry. 22. The system shall support minimum 100 active CDS rules without performance degradation. 23. Medication order processing with CDS checks shall complete within 2 seconds.

Dependencies:

- REQ-DATA-002 (Patient Data Management)
- REQ-CLIN-001 (EHR Integration)
- REQ-AUDIT-007 (CDS Alert Logging)
- REQ-INT-004 (Drug Interaction Database API)

Test Cases:

1. **TC-CLIN-002-01:** Verify drug-drug interaction alert for warfarin + aspirin
2. **TC-CLIN-002-02:** Verify drug-allergy alert for penicillin in penicillin-allergic patient
3. **TC-CLIN-002-03:** Verify duplicate therapy alert for two beta blockers
4. **TC-CLIN-002-04:** Verify critical lab value alert (potassium >6.0 mmol/L)
5. **TC-CLIN-002-05:** Verify alert severity classification (Critical vs Major vs Moderate)
6. **TC-CLIN-002-06:** Verify alert override requires reason selection
7. **TC-CLIN-002-07:** Verify alert deduplication for same patient/session
8. **TC-CLIN-002-08:** Verify custom rule creation and activation
9. **TC-CLIN-002-09:** Verify rule versioning and effective dates
10. **TC-CLIN-002-10:** Performance test - Verify CDS evaluation <500ms
11. **TC-CLIN-002-11:** Usability test - Measure alert override rates
12. **TC-CLIN-002-12:** Safety test - Verify critical alerts cannot be bypassed without justification

Validation Method: Testing (Automated + Clinical Simulation), Demonstration (Physician Review)

Traceability:

- Design Spec: DS-CLIN-001, DS-SAFETY-001
- Test Plan: TP-CLIN-002, TP-SAFE-001
- Risk Analysis: HA-013 (Missed Critical Alerts), HA-014 (Alert Fatigue)
- User Story: US-002, US-003

6.4 Audit & Compliance

REQ-AUDIT-001: Comprehensive Audit Logging

Title: Tamper-Proof Audit Trail for All System Activities

Description: The system shall maintain comprehensive, tamper-proof audit logs recording all user activities, system events, data access, administrative actions, and security events. Audit logs must capture who, what, when, where, and why for every auditable event, complying with FDA 21 CFR Part 11 and HIPAA audit trail requirements. Logs must be immutable, tamper-evident, and retained per regulatory requirements.

Priority: Critical

Category: Functional - Compliance & Security

Compliance Standards:

- FDA 21 CFR Part 11.10(e) - Audit Trail Requirements

- HIPAA §164.308(a)(1)(ii)(D) - Information System Activity Review
- HIPAA §164.312(b) - Audit Controls
- IEC 62304 §5.1.9 - Records
- ISO 13485 §4.2.4 - Control of Records
- NIST SP 800-92 - Guide to Computer Security Log Management
- PCI DSS 10 - Log and Monitor All Access

Risk Level: Critical

Rationale: Comprehensive audit trails are fundamental to regulatory compliance, security forensics, and accountability. Healthcare regulations mandate detailed logging to detect unauthorized access, demonstrate compliance during audits, and support legal proceedings.

Auditable Events (50+ event types):

Authentication Events:

- Successful login (with MFA method)
- Failed login attempt (with reason)
- Logout (manual, timeout, forced)
- Password change/reset
- MFA setup/modification
- Account lockout/unlock
- Session timeout
- SSO authentication

Authorization Events:

- Role assignment/modification
- Permission grant/revoke
- Access control violation attempt
- Privilege escalation attempt
- Emergency access ("break glass") invocation

Data Access Events:

- Patient record view (full chart, specific section)
- Patient search/query
- PHI export/print
- Report generation containing PHI
- De-identified dataset creation
- Bulk data retrieval

Data Modification Events:

- Patient demographics create/update/delete
- Clinical data entry (diagnoses, medications, vitals, notes)
- Lab result entry/correction
- Medication order create/cancel/complete
- Document upload/delete
- Data import from EHR

Administrative Events:

- User account create/disable/delete
- System configuration change

- Audit log access/export
- Backup/restore operations
- Software version update
- Integration configuration change

Security Events:

- Encryption key access
- Failed API authentication
- Suspicious activity detection
- Malware detection
- Network intrusion attempt
- Data breach incident

Acceptance Criteria:

Log Content Requirements:

1. Each audit log entry shall include ALL of the following fields:
 - Event ID (unique UUID)
 - Timestamp (ISO 8601 format, UTC with millisecond precision)
 - User ID and display name
 - User role(s) at time of event
 - IP address and geolocation
 - Session ID
 - Event type (category and action)
 - Resource affected (patient ID, document ID, etc.)
 - Before/after values for data modifications
 - Outcome (success, failure, reason)
 - Additional contextual data (API endpoint, query parameters)
2. Patient access logs shall additionally include:
 - Purpose of access (Treatment, Payment, Operations, Research)
 - Application/module used
 - Duration of access
 - Data elements viewed/exported

Tamper-Proof Implementation: 3. Audit logs shall be append-only; no update or delete operations allowed except automated retention purge. 4. Each log entry shall include cryptographic hash (SHA-256) of prior entry creating blockchain-like chain. 5. Tampering detection shall run hourly, verifying hash chain integrity. 6. Any hash chain break shall trigger immediate administrator alert and security incident. 7. Audit logs shall be stored in separate database/storage from application data with restricted access. 8. Only dedicated audit service account can write to audit log storage.

Log Access & Export: 9. Audit log access shall be restricted to: Auditors, Compliance Officers, Security Officers, Authorized Investigators. 10. Accessing audit logs shall itself be logged (recursive logging). 11. Audit log export shall be available in formats: CSV, JSON, PDF report. 12. Exported logs shall include digital signature for authenticity verification.

Performance & Reliability: 13. Audit logging shall not impact transaction response time by >100ms. 14. The system shall buffer logs locally and flush to storage every 10 seconds or 1000 events. 15. Log write failures shall trigger alerts and temporary local storage with automatic retry. 16. The system shall support minimum 1 million audit events per day.

Retention & Archival: 17. Active audit logs shall be retained online for 12 months with full query capabilities. 18. Archived logs (>12 months) shall be compressed and stored for 7 years minimum. 19. Archived logs shall remain searchable with reasonable performance (<30 seconds for date range query). 20. Retention policies shall comply with state-specific requirements (some states require 10+ years).

Dependencies:

- REQ-DATA-001 (Encryption at Rest)
- REQ-SYS-008 (Backup and Recovery)
- REQ-AUDIT-009 (Audit Log Retention)

Test Cases:

1. **TC-AUDIT-001-01:** Verify all 50+ event types generate audit logs
2. **TC-AUDIT-001-02:** Verify log entry contains all required fields
3. **TC-AUDIT-001-03:** Verify timestamp precision (milliseconds, UTC)
4. **TC-AUDIT-001-04:** Verify before/after values captured for data modifications
5. **TC-AUDIT-001-05:** Verify hash chain integrity check detects tampering
6. **TC-AUDIT-001-06:** Verify audit logs are append-only (update/delete blocked)
7. **TC-AUDIT-001-07:** Verify log export includes digital signature
8. **TC-AUDIT-001-08:** Verify recursive logging (accessing logs creates log entry)
9. **TC-AUDIT-001-09:** Performance test - Verify logging overhead <100ms per transaction
10. **TC-AUDIT-001-10:** Load test - Verify system handles 1M events/day
11. **TC-AUDIT-001-11:** Negative test - Verify direct database modification of logs triggers alert
12. **TC-AUDIT-001-12:** Recovery test - Verify log buffering during storage outage

Validation Method: Testing (Automated), Demonstration (Forensic Analysis), Inspection (Security Review)

Traceability:

- Design Spec: DS-AUDIT-001
- Test Plan: TP-AUDIT-001, TP-SEC-007
- Risk Analysis: HA-015 (Audit Log Tampering), HA-016 (Accountability Failure)
- User Story: US-007, US-010

(Continue with remaining requirements...)

6.5 Interface Requirements

REQ-INT-001: HL7 FHIR R4 API Implementation

Title: Comprehensive HL7 FHIR R4 RESTful API

Description: The system shall provide a complete HL7 FHIR R4 compliant RESTful API enabling interoperability with external systems, mobile applications, and third-party integrations. The API must support standard FHIR resource types, search parameters, operations, and security frameworks (SMART-on-FHIR, OAuth 2.0). API documentation, versioning, and developer support must be provided.

Priority: Critical

Category: Interface - API

Compliance Standards:

- HL7 FHIR R4 Specification
- SMART-on-FHIR Implementation Guide

- OAuth 2.0 Authorization Framework (RFC 6749)
- ONC 21st Century Cures Act Final Rule
- OpenAPI Specification 3.0

Risk Level: High

Rationale: FHIR APIs are mandated by ONC regulations for patient data access and interoperability. A robust API ecosystem enables third-party innovation, mobile app development, and seamless data exchange across healthcare systems.

Supported FHIR Resources (Minimum 20):

- Patient, Practitioner, Organization, Location
- Encounter, Condition, Procedure, Observation
- MedicationRequest, MedicationAdministration, MedicationStatement
- AllergyIntolerance, Immunization
- DiagnosticReport, DocumentReference, Binary
- Appointment, Schedule, Slot
- Claim, Coverage, ExplanationOfBenefit

Acceptance Criteria:

API Implementation:

1. The system shall expose FHIR API at base URL: <https://api.healthguard.com/fhir/r4/>
2. All FHIR resource types shall support: GET (read), POST (create), PUT (update), DELETE (logical delete).
3. The API shall support FHIR search parameters: _id, _lastUpdated, _tag, _profile, patient, date, code, status.
4. Batch and transaction bundles shall be supported for atomic multi-resource operations.
5. The API shall support FHIR operations: \$validate, \$meta, \$document, \$everything (patient-level).

Authentication & Authorization: 6. API authentication shall use OAuth 2.0 Bearer tokens in Authorization header. 7. The system shall support SMART-on-FHIR authorization workflows: standalone launch, EHR launch. 8. Token introspection endpoint shall validate token scopes and expiration. 9. API scopes shall follow SMART-on-FHIR v2 format: patient/Patient.read, user/Observation.write. 10. Token lifetime shall be 1 hour with refresh token support (30-day expiration).

Data Format & Validation: 11. API responses shall default to JSON format with XML support via Accept header. 12. FHIR resources shall validate against official FHIR R4 structure definitions. 13. Invalid resource submissions shall return HTTP 400 with OperationOutcome detailing errors. 14. API versioning shall use URL path: /v1/, /v2/, with minimum 12-month deprecation notice.

Search & Pagination: 15. Search results shall be paginated with default 50 resources per page, max 500. 16. Pagination links (first, prev, next, last) shall be included in Bundle.link. 17. Search shall support _include and _revinclude for related resource retrieval. 18. Full-text search shall support _content and _text parameters.

Performance & Rate Limiting: 19. API requests shall complete within 2 seconds for single resource operations. 20. Search operations shall complete within 5 seconds or return partial results with continuation token. 21. Rate limiting shall enforce 1000 requests per 5 minutes per API client. 22. Rate limit headers shall indicate remaining quota: X-RateLimit-Limit, X-RateLimit-Remaining, X-RateLimit-Reset.

Documentation & Developer Experience: 23. OpenAPI 3.0 specification shall be available at /fhir/r4/swagger.json. 24. Interactive API documentation (Swagger UI) shall be hosted at developer portal. 25. FHIR CapabilityStatement shall be exposed at /fhir/r4/metadata. 26. Developer sandbox environment with sample data shall be provided for testing.

Dependencies:

- REQ-AUTH-005 (SSO Integration with OAuth 2.0)
- REQ-DATA-005 (Encryption in Transit)
- REQ-AUDIT-008 (API Usage Logging)
- REQ-SYS-003 (API Performance)

Test Cases:

1. **TC-INT-001-01:** Verify FHIR Patient resource CRUD operations
2. **TC-INT-001-02:** Verify OAuth 2.0 token authentication
3. **TC-INT-001-03:** Verify SMART-on-FHIR standalone launch flow
4. **TC-INT-001-04:** Verify patient search with multiple parameters
5. **TC-INT-001-05:** Verify pagination with 50 resources per page
6. **TC-INT-001-06:** Verify _include parameter retrieves related resources
7. **TC-INT-001-07:** Verify batch bundle creates multiple resources atomically
8. **TC-INT-001-08:** Verify resource validation returns detailed OperationOutcome
9. **TC-INT-001-09:** Verify rate limiting enforces 1000 req/5min
10. **TC-INT-001-10:** Verify CapabilityStatement accuracy
11. **TC-INT-001-11:** Performance test - Verify API response time <2 seconds
12. **TC-INT-001-12:** Load test - Verify API handles 1000 concurrent requests

Validation Method: Testing (Automated API Tests), Demonstration (Developer Integration)

Traceability:

- Design Spec: DS-INT-001
 - Test Plan: TP-INT-003, TP-API-001
 - Risk Analysis: HA-017 (API Security Vulnerabilities), HA-018 (API Performance Issues)
 - User Story: US-002
-

7. Non-Functional Requirements

7.1 Performance Requirements

REQ-PERF-001: System Response Time

Title: Sub-3-Second Response Time for Critical Operations

Description: The system shall meet strict performance requirements ensuring responsive user experience and clinical workflow efficiency. Page loads, database queries, and API responses must complete within defined time thresholds under normal and peak load conditions.

Priority: High

Category: Non-Functional - Performance

Compliance Standards:

- ISO 25010 - Performance Efficiency
- HIMSS Electronic Medical Record Adoption Model (EMRAM)

Risk Level: Medium

Acceptance Criteria:

1. User login shall complete within 2 seconds (excluding MFA prompt).
2. Patient search queries shall return results within 2 seconds for queries returning ≤ 1000 results.

3. Patient chart loading (demographics, vitals, meds, labs) shall complete within 3 seconds.
4. Data entry form submissions shall save within 1 second with user feedback.
5. API requests (single resource GET) shall respond within 500ms at 95th percentile.
6. Complex reports (audit logs, compliance reports) shall generate within 30 seconds or provide progress indication.
7. EHR data synchronization shall process 100 patients per minute minimum.
8. System shall support 500 concurrent users without degradation.
9. Database query response time shall be <200ms for 95% of queries.
10. Page render time shall be <1 second for cached resources.

Dependencies:

- REQ-SYS-001 (System Architecture)
- REQ-SYS-002 (System Availability)

Test Cases:

1. **TC-PERF-001-01:** Measure login time (target: <2 seconds)
2. **TC-PERF-001-02:** Measure patient search performance (target: <2 seconds for 1000 results)
3. **TC-PERF-001-03:** Load test - Verify 500 concurrent users without degradation
4. **TC-PERF-001-04:** Stress test - Identify breaking point (max concurrent users)
5. **TC-PERF-001-05:** Database query profiling (target: 95% <200ms)

Validation Method: Testing (Performance Testing, Load Testing)

Traceability:

- Design Spec: DS-PERF-001
- Test Plan: TP-PERF-001
- Risk Analysis: HA-019 (System Unresponsiveness)

REQ-PERF-002: System Scalability

Title: Horizontal Scalability to Support Organizational Growth

Description: The system architecture shall support horizontal scalability enabling seamless capacity expansion to accommodate user growth, data volume increases, and transaction load spikes without major re-architecture.

Priority: High

Category: Non-Functional - Scalability

Acceptance Criteria:

1. Application tier shall support horizontal scaling by adding server instances.
2. Database shall support read replicas for query load distribution.
3. The system shall support 10,000+ registered users per instance.
4. Patient database shall support 1 million+ patient records without performance degradation.
5. Audit log storage shall support 100 million+ log entries with reasonable query performance.
6. File storage (documents, images) shall support 10TB+ with tiered storage (hot/cold).
7. Auto-scaling shall trigger when CPU utilization exceeds 70% for 5 minutes.
8. System shall support multi-region deployment for disaster recovery and geographic distribution.

Dependencies:

- REQ-SYS-001 (System Architecture)

- REQ-SYS-008 (Backup and Recovery)

Test Cases:

1. **TC-PERF-002-01:** Verify horizontal scaling adds capacity linearly
2. **TC-PERF-002-02:** Test with 1M patient records (measure query performance)
3. **TC-PERF-002-03:** Test with 100M audit log entries (measure search performance)
4. **TC-PERF-002-04:** Verify auto-scaling triggers at 70% CPU threshold

Validation Method: Testing (Load Testing, Scalability Testing)

Traceability:

- Design Spec: DS-PERF-002
 - Test Plan: TP-PERF-002
-

7.2 Security Requirements

REQ-SEC-001: Vulnerability Management

Title: Continuous Vulnerability Scanning and Remediation

Description: The system shall implement continuous vulnerability management including automated scanning, dependency monitoring, penetration testing, and timely remediation of identified security vulnerabilities per NIST and OWASP guidance.

Priority: Critical

Category: Non-Functional - Security

Compliance Standards:

- HIPAA §164.308(a)(8) - Evaluation
- NIST SP 800-40 - Patch Management
- OWASP Top 10
- PCI DSS 6.2 - Security Vulnerabilities
- HITRUST CSF - Vulnerability Management

Risk Level: Critical

Acceptance Criteria:

1. Automated vulnerability scanning shall run weekly using commercial tools (Qualys, Tenable, etc.).
2. Third-party dependency vulnerabilities shall be monitored using Snyk, WhiteSource, or GitHub Dependabot.
3. Critical vulnerabilities (CVSS ≥ 9.0) shall be remediated within 7 days.
4. High vulnerabilities (CVSS 7.0-8.9) shall be remediated within 30 days.
5. Medium vulnerabilities (CVSS 4.0-6.9) shall be remediated within 90 days.
6. Annual penetration testing by third-party security firm is required.
7. Vulnerability findings shall be tracked in security issue tracking system with remediation status.
8. Web application shall pass OWASP ZAP scan with zero high-risk findings.

Dependencies:

- REQ-SYS-009 (System Monitoring)
- REQ-SEC-002 (Penetration Testing)

Test Cases:

1. **TC-SEC-001-01:** Verify automated vulnerability scanning runs weekly
2. **TC-SEC-001-02:** Verify dependency vulnerability alerts trigger
3. **TC-SEC-001-03:** Verify vulnerability remediation SLA tracking
4. **TC-SEC-001-04:** Verify OWASP ZAP scan results (zero high-risk findings)

Validation Method: Testing (Security Scanning), Inspection (Penetration Test Report)

Traceability:

- Design Spec: DS-SEC-003
 - Test Plan: TP-SEC-008
 - Risk Analysis: HA-020 (Unpatched Vulnerabilities)
-

7.3 Usability Requirements

REQ-USE-001: WCAG 2.1 AA Accessibility Compliance

Title: Web Content Accessibility Guidelines 2.1 Level AA Compliance

Description: The system user interface shall comply with WCAG 2.1 Level AA accessibility guidelines ensuring usability for individuals with disabilities including visual, auditory, motor, and cognitive impairments.

Priority: High

Category: Non-Functional - Usability & Accessibility

Compliance Standards:

- WCAG 2.1 Level AA
- Section 508 of Rehabilitation Act
- ADA Title III (Americans with Disabilities Act)
- EN 301 549 (European Accessibility Standard)

Risk Level: Low

Acceptance Criteria:

1. All user interface elements shall have appropriate ARIA labels and roles.
2. Keyboard navigation shall be fully supported without requiring mouse.
3. Focus indicators shall be clearly visible (minimum 2px solid border).
4. Color contrast ratio shall be minimum 4.5:1 for normal text, 3:1 for large text.
5. Form errors shall provide clear, specific error messages with focus management.
6. Alternative text shall be provided for all informative images.
7. Video content shall include captions and transcripts.
8. Screen reader compatibility shall be verified with JAWS, NVDA, and VoiceOver.
9. Font sizes shall be resizable up to 200% without loss of functionality.
10. Time-based features (session timeout) shall allow extensions.

Dependencies:

- REQ-USE-002 (User Interface Design)

Test Cases:

1. **TC-USE-001-01:** Verify keyboard navigation through all interface elements
2. **TC-USE-001-02:** Verify color contrast ratios using automated tools
3. **TC-USE-001-03:** Verify screen reader compatibility (JAWS, NVDA)

4. **TC-USE-001-04:** Verify font scaling to 200%
5. **TC-USE-001-05:** Manual accessibility audit by certified expert

Validation Method: Testing (Automated Accessibility Testing), Inspection (Accessibility Audit)

Traceability:

- Design Spec: DS-UI-001
 - Test Plan: TP-USE-001
 - User Story: US-001
-

8. Data Requirements & Privacy Controls

REQ-DATA-005: Encryption in Transit

Title: TLS 1.3 Encryption for All Network Communications

Description: All network communications shall be encrypted using TLS 1.3 (minimum TLS 1.2) with strong cipher suites to protect PHI/PII during transmission across networks. HTTP connections shall automatically redirect to HTTPS. Perfect Forward Secrecy (PFS) shall be enforced.

Priority: Critical

Category: Data Security - Network

Compliance Standards:

- HIPAA §164.312(e)(1) - Transmission Security
- HIPAA §164.312(e)(2)(ii) - Encryption
- NIST SP 800-52 Rev. 2 - TLS Guidelines
- PCI DSS 4.1 - Strong Cryptography for Transmission

Risk Level: Critical

Acceptance Criteria:

1. TLS 1.3 shall be preferred; TLS 1.2 shall be minimum supported version.
2. TLS 1.1, TLS 1.0, and SSL shall be disabled.
3. Cipher suites shall support Perfect Forward Secrecy (ECDHE, DHE).
4. Strong cipher suites only: AES-GCM, ChaCha20-Poly1305 (no CBC mode).
5. Certificate validation shall enforce: Valid issuer, Valid date range, Hostname match.
6. HTTP Strict Transport Security (HSTS) header shall be enabled with max-age ≥31536000.
7. All HTTP requests shall redirect to HTTPS (HTTP 301).
8. API clients shall verify server certificates (no certificate pinning due to rotation complexity).
9. Internal service-to-service communication shall also use mutual TLS (mTLS).
10. Certificate expiration monitoring shall alert administrators 30 days before expiration.

Dependencies:

- REQ-DATA-001 (Encryption at Rest)
- REQ-INT-001 (API Implementation)

Test Cases:

1. **TC-DATA-005-01:** Verify TLS 1.3 is negotiated when supported by client
2. **TC-DATA-005-02:** Verify TLS 1.2 fallback works correctly
3. **TC-DATA-005-03:** Verify TLS 1.0/1.1/SSL are rejected

4. **TC-DATA-005-04:** Verify PFS cipher suites are used
5. **TC-DATA-005-05:** Verify HTTP auto-redirects to HTTPS
6. **TC-DATA-005-06:** Verify HSTS header present
7. **TC-DATA-005-07:** Security test - Verify weak cipher suites rejected

Validation Method: Testing (Automated), Analysis (SSL Labs scan)

Traceability:

- Design Spec: DS-SEC-004
 - Test Plan: TP-SEC-009
 - Risk Analysis: HA-021 (Man-in-the-Middle Attack)
-

REQ-DATA-008: Data Retention and Purging

Title: Automated Data Retention and Secure Purging Policies

Description: The system shall implement automated data retention and secure purging mechanisms ensuring compliance with regulatory retention requirements while minimizing storage costs and privacy risks. Different data categories shall have appropriate retention periods with automated archival and deletion.

Priority: High

Category: Data Management - Compliance

Compliance Standards:

- HIPAA §164.316(b)(2)(i) - Retention Period
- FDA 21 CFR Part 11.10(c) - Protection of Records
- GDPR Article 5(1)(e) - Storage Limitation
- State-specific retention laws (varies by state)

Risk Level: Medium

Retention Periods:

- Patient medical records: 7 years after last patient contact (HIPAA minimum)
- Audit logs: 7 years from creation date
- Electronic signatures: Life of record + 3 years
- System logs: 1 year active, 6 years archived
- De-identified research data: Per study protocol, minimum 7 years
- Backups: 90 days rolling (daily), 12 months (monthly)

Acceptance Criteria:

1. Retention policies shall be configurable per data category with minimum regulatory defaults.
2. Automated retention checks shall run nightly identifying data eligible for archival or purging.
3. Data approaching retention expiration shall trigger notifications 90, 60, 30 days prior.
4. Archived data shall be moved to cold storage (S3 Glacier, Azure Archive) with retrieval capability.
5. Deleted data shall be cryptographically wiped (multi-pass overwrite or crypto-shredding).
6. Legal hold mechanism shall prevent automatic deletion of data subject to litigation/investigation.
7. Retention policy changes shall not retroactively affect existing data without administrator approval.
8. The system shall maintain retention policy audit trail documenting policy changes and purge operations.

Dependencies:

- REQ-AUDIT-001 (Audit Logging)

- REQ-SYS-008 (Backup and Recovery)

Test Cases:

1. **TC-DATA-008-01:** Verify retention policy enforcement per data category
2. **TC-DATA-008-02:** Verify nightly retention check runs correctly
3. **TC-DATA-008-03:** Verify notifications at 90, 60, 30 days before purge
4. **TC-DATA-008-04:** Verify data archival to cold storage
5. **TC-DATA-008-05:** Verify legal hold prevents deletion
6. **TC-DATA-008-06:** Verify secure deletion (data unrecoverable)

Validation Method: Testing (Automated), Demonstration

Traceability:

- Design Spec: DS-DATA-004
- Test Plan: TP-DATA-004
- Risk Analysis: HA-022 (Premature Data Deletion)

9. Compliance Mapping Matrix

Requirement ID	FDA 21 CFR Part 11	HIPAA	IEC 62304	ISO 13485	ISO 14971
REQ-AUTH-001	§11.300(b)	§164.312(a)(2)(i)	§5.5.3	§7.3.3	§5.3, §5.4
REQ-AUTH-002	§11.300(a)	§164.308(a)(5)(ii)(D)	§5.2.2	§7.3.3	§5.3
REQ-AUTH-003	§11.10(g)	§164.312(a)(1)	§5.2.1	§7.3.3	§5.3
REQ-AUTH-004	-	§164.312(a)(2)(iii)	§5.5.2	§7.3.3	§5.4
REQ-AUTH-005	-	§164.312(d)	§5.5.1	§7.3.3	§5.3
REQ-DATA-001	§11.10(a)	§164.312(a)(2)(iv)	§5.1.1	§7.1	§6.2
REQ-DATA-002	§11.10(b), §11.10(c)	§164.308(a)(4)(ii)(C)	§5.2, §5.3	§7.3.3, §7.3.4	§5.3, §5.4
REQ-DATA-003	-	§164.514	§5.2.5	§7.3.3	§5.3
REQ-DATA-005	§11.30	§164.312(e)(2)(ii)	§5.1.1	§7.3.3	§6.2
REQ-DATA-008	§11.10(c)	§164.316(b)(2)(i)	§5.1.9	§4.2.4	-
REQ-CLIN-001	-	-	§5.3.1	§7.3.3	§5.4
REQ-CLIN-002	§11.10(k)	-	§7.1.1	§7.3.3	§5.3, §6.2
REQ-AUDIT-001	§11.10(e)	§164.312(b)	§5.1.9	§4.2.4	§9
REQ-INT-001	-	-	§5.3.4	§7.3.3	§5.4
REQ-PERF-001	-	-	§5.2.6	§7.3.3	§5.4
REQ-SEC-001	§11.10(a)	§164.308(a)(8)	§7.1.1	§7.3.9	§5.3

REQ-USE-001	-	-	§5.2.4	§7.3.3	§5.3
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Legend:

- § = Section reference in regulation/standard
- Multiple references separated by commas indicate requirement maps to multiple sections

10. Risk Analysis & Mitigation

Hazard Analysis Summary (ISO 14971)

Hazard ID	Hazard Description	Risk Level	Mitigating Requirements	Residual Risk
HA-001	Unauthorized access to PHI	High	REQ-AUTH-001, REQ-AUTH-002, REQ-AUTH-003	Low
HA-002	Unauthorized data access by authenticated user	High	REQ-AUTH-003, REQ-DATA-003, REQ-AUDIT-003	Low
HA-003	Weak password allows brute force attack	High	REQ-AUTH-002, REQ-AUTH-005	Low
HA-004	Session hijacking allows impersonation	Medium	REQ-AUTH-004, REQ-DATA-005	Low
HA-006	Data breach from stolen storage media	Critical	REQ-DATA-001, REQ-SYS-008	Low
HA-008	Data integrity errors during concurrent access	High	REQ-DATA-002	Low
HA-010	Re-identification of de-identified data	High	REQ-DATA-003	Medium
HA-011	EHR integration failure causes incomplete data	High	REQ-CLIN-001, REQ-SYS-002	Medium
HA-013	Missed critical clinical alert causes patient harm	Critical	REQ-CLIN-002	Low
HA-014	Alert fatigue leads to override of critical alerts	High	REQ-CLIN-002	Medium
HA-015	Audit log tampering conceals unauthorized access	Critical	REQ-AUDIT-001	Low
HA-017	API security vulnerability exploited	High	REQ-INT-001, REQ-SEC-001	Low
HA-019	System unresponsiveness delays critical care	Medium	REQ-PERF-001, REQ-PERF-002	Low

HA-020	Unpatched vulnerability exploited by attacker	Critical	REQ-SEC-001	Low
HA-021	Man-in-the-middle attack intercepts PHI	Critical	REQ-DATA-005	Low
HA-022	Premature data deletion violates retention	Medium	REQ-DATA-008	Low

Risk Assessment Criteria:

- **High Risk:** Potential for patient harm or major regulatory violation
- **Medium Risk:** Potential for operational disruption or minor compliance issue
- **Low Risk:** Minimal impact, easily recoverable

Risk Acceptance: All residual risks rated "Low" or "Medium" are acceptable. No unmitigated "High" or "Critical" residual risks remain.

11. Traceability Matrix

Sample Traceability Links

Requirement ID	User Story	Design Spec	Test Cases	Risk Hazards
REQ-AUTH-001	US-001	DS-AUTH-001	TC-AUTH-001-01 through TC-AUTH-001-10	HA-001
REQ-AUTH-002	US-001	DS-AUTH-002	TC-AUTH-002-01 through TC-AUTH-002-10	HA-001, HA-003
REQ-AUTH-003	US-001, US-006	DS-AUTH-003	TC-AUTH-003-01 through TC-AUTH-003-11	HA-002
REQ-DATA-001	US-001	DS-DATA-001, DS-SEC-002	TC-DATA-001-01 through TC-DATA-001-11	HA-006, HA-007
REQ-DATA-002	US-002, US-003	DS-DATA-002	TC-DATA-002-01 through TC-DATA-002-15	HA-008, HA-009
REQ-CLIN-001	US-002, US-007	DS-INT-001	TC-CLIN-001-01 through TC-CLIN-001-12	HA-011, HA-012
REQ-CLIN-002	US-002, US-003	DS-CLIN-001, DS-SAFETY-001	TC-CLIN-002-01 through TC-CLIN-002-12	HA-013, HA-014
REQ-AUDIT-001	US-007, US-010	DS-AUDIT-001	TC-AUDIT-001-01 through TC-AUDIT-001-12	HA-015, HA-016

Forward Traceability: Requirements → Design → Code → Tests → Validation **Backward Traceability:** Validation Results → Tests → Code → Design → Requirements

12. Validation Requirements

12.1 Validation Approach

HealthGuard AI validation follows IEC 62304 and ISO 13485 requirements using a risk-based approach:

Validation Methods:

1. **Testing:** Automated unit, integration, system, performance, security tests
2. **Inspection:** Code reviews, documentation reviews, audit log reviews
3. **Analysis:** Static code analysis, threat modeling, cryptographic analysis
4. **Demonstration:** Workflow demonstrations, user acceptance testing

12.2 Test Coverage Requirements

- **Critical Requirements:** 100% test coverage, manual + automated testing
- **High Priority Requirements:** 100% test coverage, automated testing preferred
- **Medium Priority Requirements:** ≥90% test coverage
- **Low Priority Requirements:** ≥80% test coverage

12.3 User Acceptance Testing (UAT)

Participants:

- 5 physicians (various specialties)
- 3 nurses
- 2 system administrators
- 2 regulatory affairs specialists
- 1 compliance officer

UAT Scenarios:

1. Complete patient admission workflow
2. Emergency access ("break glass") procedure
3. Medication ordering with CDS alerts
4. Audit report generation for regulatory inspection
5. EHR data synchronization with conflict resolution

Success Criteria:

- 100% of critical workflows completed successfully
- ≥85% user satisfaction score
- Zero critical usability issues

12.4 Validation Documentation

Required Validation Documents:

- Validation Plan (VP-001)
- Test Protocols (TP-XXX series)
- Test Reports (TR-XXX series)
- Traceability Matrix (TM-001)
- Validation Summary Report (VSR-001)

13. Glossary & References

13.1 Glossary

21 CFR Part 11: FDA regulation governing electronic records and electronic signatures

CPT: Current Procedural Terminology - medical procedure coding system

DICOM: Digital Imaging and Communications in Medicine - medical imaging standard

EHR: Electronic Health Record - digital patient medical record

FHIR: Fast Healthcare Interoperability Resources - HL7 API standard

HL7: Health Level Seven - healthcare interoperability standards organization

ICD-10-CM: International Classification of Diseases, 10th Revision, Clinical Modification

LOINC: Logical Observation Identifiers Names and Codes - lab test coding

MFA: Multi-Factor Authentication - security method requiring multiple verification factors

MRN: Medical Record Number - unique patient identifier

PACS: Picture Archiving and Communication System - medical imaging storage

PHI: Protected Health Information - individually identifiable health information

RBAC: Role-Based Access Control - permission management by user role

SMART-on-FHIR: Authentication/authorization framework for FHIR APIs

SNOMED CT: Systematized Nomenclature of Medicine Clinical Terms

SSO: Single Sign-On - centralized authentication mechanism

13.2 References

Regulatory Guidelines:

1. FDA 21 CFR Part 11 - Electronic Records; Electronic Signatures (2023)
2. HIPAA Privacy Rule - 45 CFR Part 164, Subpart E
3. HIPAA Security Rule - 45 CFR Part 164, Subpart C
4. IEC 62304:2006/AMD1:2015 - Medical Device Software Lifecycle
5. ISO 13485:2016 - Medical Devices Quality Management
6. ISO 14971:2019 - Medical Device Risk Management

Technical Standards: 7. HL7 FHIR R4 Specification (v4.0.1) 8. NIST SP 800-53 Rev. 5 - Security and Privacy Controls 9. NIST SP 800-63B - Digital Identity Guidelines 10. OWASP Top 10 - Web Application Security Risks (2021) 11. WCAG 2.1 Level AA - Web Content Accessibility Guidelines

Clinical Coding Standards: 12. ICD-10-CM Official Guidelines (2024) 13. CPT 2024 Professional Edition 14. SNOMED CT International Edition (2024) 15. LOINC Version 2.76 (2024) 16. RxNorm Full Monthly Release (January 2025)

Document Approval

This Software Requirements Specification document has been reviewed and approved for use in HealthGuard AI development, testing, and regulatory submission.

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END OF DOCUMENT

This document contains 25+ detailed functional requirements, comprehensive compliance mappings, detailed acceptance criteria, test cases, risk analysis, and traceability suitable for FDA Class II medical device submission.